Maheer Helal

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Education

- **B.Sc in Computer Science and Engineering** *Feb 2021 – Jan 2025*
  BRAC University Dhaka, Bangladesh
  - CGPA: 3.85/4.00 (Highest Distinction)

Research Interests

Natural Language Processing (NLP), Large Language Models (LLMs), Clinical NLP, AI for healthcare, Multimodal Learning

Teaching Experience

- **Adjunct Lecturer** *Jun 2025 – Present*
  BRAC University Dhaka, Bangladesh
  - **Assigned courses:** Algorithms (CSE 221), Digital Logic Design (CSE 260), Artificial Intelligence (CSE 422), Operating Systems (CSE 321), Microprocessors (CSE 341), Machine Learning (CSE 427), Introduction to Robotics (CSE 461), Introduction to Computer Science (CSE 101/103).
  - **AI & Machine Learning:** Delivered lectures on classical AI and basic machine learning algorithms; supervised student ML projects from requirement analysis through experimentation, demonstration, and viva. In the Machine Learning lab, students ran their own experiments and reported them as IEEE-format papers, which were evaluated for academic rigor, data sourcing, and depth of analysis.
  - **Hardware & Systems:** Facilitated hands-on labs in Digital Logic Design, Microprocessors, and Robotics (trainer boards, ICs, EMU 8086 Assembly, sensors, Raspberry Pi, actuators); supervised student-built systems, including a semester-long project to build a portable, untethered robot with a real-world use case.

- **Undergraduate Teaching Assistant** *Sep 2023 – Jan 2025*
  BRAC University Dhaka, Bangladesh
  - Assigned course: Data Structures (CSE 220).
  - Guided students in weekly labs; clarified issues and debugging.
  - Graded assignments and quizzes, ensuring timely feedback.

Research Projects

- **EEG and Eye Tracking Feature Contextualization based Affective Computing through LightGBM enabled Stacked Ensembling of LLMs** *Undergraduate Thesis*
  BRAC University
  - Designed a custom text-encoding pipeline to transform multi-channel EEG and eye-tracking data into structured text sequences.
  - Used FLAN-T5, GPT-2, and BERT on these encoded multimodal inputs for emotion classification.
  - Implemented a LightGBM-based stacked ensemble to integrate model outputs for robust affective analysis.
- **LLM Based Bangla Sign Language Translation with Mixture of Experts(MoE) Vision Encoders** *Ongoing*
  - Benchmarking Meta's MoE-ViE vision encoder on Bangla sign language video via zero-shot video and text retrieval with Bangla sentences.
  - Analyzed frame-averaged embeddings, finding them dominated by signer and session appearance rather than signing content, motivating temporal modeling over frozen MoE features.

English Language Proficiency

IELTS Academic *October 2026*

Listening	Reading	Writing	Speaking	Overall
8.5	7.5	6.5	7.5	7.5

Honors and Awards

- **Graduated with Highest Distinction** *2025*
  BRAC University
  - Awarded to undergraduate students with a CGPA of 3.8 or above.
- **Dean's & VC's List** — appeared 8 times
  BRAC University
  - Awarded to undergraduate students with a CGPA of 3.7 or above while completing a full 12-credit load.
- **Merit Scholarship for Academic Excellence**
  BRAC University
  - Awarded to students completing 12 credits per semester with no retakes, repeats, or withdrawals, and maintaining a CGPA of 3.7 or higher.
- **President's Scout Award** *2016*
  Bangladesh Scouts
  - Highest national honor awarded by Bangladesh Scouts for excellence in leadership, community service, and scouting skills.

Courses and Certifications

- [🔗] **AI Fundamentals** *DataCamp*
- [🔗] **Large Language Models (LLMs) Concepts** *DataCamp*
- [🔗] **Biomedical Image Analysis in Python** *DataCamp*
- [🔗] **Understanding Artificial Intelligence** *DataCamp*

Skills

- **Programming Languages:** Python, Java, SQL, HTML, CSS, LaTeX
- **Tools for Deep Learning:** PyTorch (CUDA), TensorFlow, Transformers (Hugging Face), BitsandBytes
- **LLM Tooling and Evaluation:** Hugging Face Datasets, Hugging Face Evaluate, Scikit-learn
- **Data Science and Analytics:** Pandas, NumPy, SciPy, Matplotlib, Seaborn, OpenCV
- **Version Control and Environment:** Git, GitHub

Projects

- [🐙] **Medical Chat Summarization using BioBART** *(Python, TensorFlow, Hugging Face Transformers)*
  - Investigated biomedical domain-specific transformers for clinical NLP by fine-tuning BioBART on the UIU Medical Dialogue Dataset to generate structured SOAP notes from multi-turn doctor-patient dialogues. Achieved ROUGE-1,2,L scores of 0.67,0.40,0.48 respectively.
- [🐙] **Image Segmentation Using UNET** *(Python, TensorFlow, scikit-learn, pandas, numpy)*
  - Implemented a U-Net–based deep learning model for semantic segmentation of plant disease images, achieving accurate pixel-level mask predictions under varied real-world conditions. Model achieved 90% accuracy in plant disease detection.
- [🐙] **Money Management App** *(Django)*
  - Built a personal-finance platform with secure email-verified accounts, bill/expense tracking with automated due-date reminders, and real-time stock-portfolio analytics (yfinance + Plotly).

References

1. **Md. Golam Rabiul Alam**
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   BRAC University
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2. **Rafeed Rahman**
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